

Retail Sales

Analysis Using SQL

Question

Q1) Write a SQL query to retrieve all columns for sales made on '2022-11-05':

```
SELECT *
FROM retail_sales
WHERE sale_date = '2022-11-05';
```

	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
▶	180	2022-11-05	10:47:00	117	Male	41	Clothing	3	300.00	129.00	900.00
	214	2022-11-05	16:31:00	53	Male	20	Beauty	2	30.00	8.10	60.00
	240	2022-11-05	11:49:00	95	Female	23	Beauty	1	300.00	123.00	300.00
	856	2022-11-05	17:43:00	102	Male	54	Electronics	4	30.00	9.30	120.00
	943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300.00	318.00	1200.00
	1137	2022-11-05	22:34:00	104	Male	46	Beauty	2	500.00	145.00	1000.00
	1256	2022-11-05	09:58:00	29	Male	23	Clothing	2	500.00	190.00	1000.00
	1265	2022-11-05	14:35:00	86	Male	55	Clothing	3	300.00	111.00	900.00
	1587	2022-11-05	20:06:00	140	Female	40	Beauty	4	300.00	105.00	1200.00
	1819	2022-11-05	20:44:00	83	Female	35	Beauty	2	50.00	13.50	100.00
	1896	2022-11-05	20:19:00	87	Female	30	Electronics	2	25.00	30.75	50.00

Q2) Write a SQL query to retrieve all transactions where the category is 'Clothing' and the quantity sold is more than 4 in the month of Nov-2022:

```
SELECT *
FROM retail_sales
WHERE category = 'Clothing'
AND quantity >= 4
AND DATE_FORMAT(sale_date, '%Y-%m') = '2022-11';
```

	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
▶	64	2022-11-15	06:34:00	7	Male	49	Clothing	4	25.00	8.50	100.00
	146	2022-11-10	22:01:00	74	Male	38	Clothing	4	50.00	49.00	200.00
	159	2022-11-10	21:30:00	42	Male	26	Clothing	4	50.00	23.50	200.00
	284	2022-11-12	09:17:00	129	Male	43	Clothing	4	50.00	20.50	200.00
	547	2022-11-14	07:36:00	3	Male	63	Clothing	4	500.00	250.00	2000.00
	699	2022-11-21	22:21:00	129	Female	37	Clothing	4	30.00	16.20	120.00
	735	2022-11-26	21:38:00	153	Female	64	Clothing	4	500.00	515.00	2000.00
	943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300.00	318.00	1200.00
	965	2022-11-27	21:45:00	84	Male	22	Clothing	4	50.00	13.00	200.00
	1259	2022-11-03	17:31:00	105	Female	45	Clothing	4	50.00	21.00	200.00
	1296	2022-11-26	20:42:00	45	Female	22	Clothing	4	300.00	342.00	1200.00
	1476	2022-11-11	22:27:00	130	Female	27	Clothing	4	500.00	555.00	2000.00
	1484	2022-11-23	09:29:00	22	Female	19	Clothing	4	300.00	147.00	1200.00
	1497	2022-11-19	21:44:00	109	Male	41	Clothing	4	30.00	32.40	120.00
	1615	2022-11-17	13:43:00	82	Female	61	Clothing	4	25.00	13.50	100.00
	1696	2022-11-21	17:59:00	24	Female	50	Clothing	4	50.00	55.00	200.00
	1885	2022-11-09	07:32:00	148	Female	52	Clothing	4	30.00	10.80	120.00

Q3) Write a SQL query to calculate the total sales (total_sale) for each category.

```
SELECT
    SUM(total_sale) AS net_sales, category
FROM
    retail_sales
GROUP BY category
ORDER BY category DESC;
```

net_sales	category
311445.00	Electronics
309995.00	Clothing
286790.00	Beauty

Q4) Write a SQL query to find the average age of customers who purchased items from the 'Beauty' category.

```
select avg (age ) as avg_age
from retail_sales
where category= 'Beauty';
```

avg_age
40.4157

Q 5) Write a SQL query to find all transactions where the total_sale is greater than 1000.

```
SELECT * FROM retail_sales
WHERE total_sale > 1000;
```

transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
13	2023-02-08	17:43:00	106	Male	22	Electronics	3	500.00	245.00	1500.00
15	2022-07-01	11:50:00	75	Female	42	Electronics	4	500.00	210.00	2000.00
16	2022-06-25	10:33:00	82	Male	19	Clothing	3	500.00	180.00	1500.00
31	2023-12-31	17:47:00	3	Male	44	Electronics	4	300.00	129.00	1200.00
46	2022-11-08	17:50:00	54	Female	20	Electronics	4	300.00	84.00	1200.00
47	2022-10-22	17:22:00	96	Female	40	Beauty	3	500.00	600.00	1500.00
54	2022-10-20	10:17:00	142	Female	38	Electronics	3	500.00	200.00	1500.00
58	2023-09-16	19:18:00	53	Male	18	Clothing	4	300.00	75.00	1200.00
65	2022-12-11	20:03:00	84	Male	51	Electronics	4	500.00	160.00	2000.00
67	2023-08-19	20:19:00	119	Female	48	Beauty	4	300.00	129.00	1200.00
72	2023-12-06	19:19:00	5	Female	20	Electronics	4	500.00	195.00	2000.00
74	2023-10-05	19:50:00	56	Female	18	Beauty	4	500.00	205.00	2000.00
78	2023-02-17	21:08:00	68	Female	47	Clothing	3	500.00	265.00	1500.00
89	2023-12-30	21:15:00	117	Female	55	Electronics	4	500.00	590.00	2000.00

Q6) Write a SQL query to find the total number of transactions (transaction_id) made by each gender in each category.

```
SELECT COUNT(transactions_id) AS total_transactions,
    gender,
    category
FROM retail_sales
GROUP BY gender, category;
```

total_transactions	gender	category
281	Male	Beauty
347	Female	Clothing
343	Male	Electronics
351	Male	Clothing
330	Female	Beauty
335	Female	Electronics

Q7) Write a SQL query to calculate the average sale for each month. Find out best selling month in each year:

```
SELECT
    AVG(total_sale) AS avg_sales
    YEAR(sale_date) AS year,
    MONTH(sale_date) AS month
FROM retail_sales
GROUP BY 2, 3;
```

avg_sales	year	month
460.769231	2022	12
481.395349	2022	6
495.964912	2023	8
462.739726	2023	9
453.452381	2023	11
427.678571	2023	7
467.137931	2022	10
366.136364	2022	2
535.531915	2023	2
488.888888	2023	5

Q8) Write a SQL query to find the top 5 customers based on the highest total sales

```
SELECT
    customer_id,
    SUM(total_sale) as total_sales
FROM retail_sales
GROUP BY 1
ORDER BY 1,2 DESC
limit 5;
```

customer_id	total_sales
1	30750.00
2	25295.00
3	38440.00
4	23580.00
5	30405.00

Q9) Write a SQL query to find the number of unique customers who purchased items from each category.

```
SELECT
    category,
    COUNT(DISTINCT customer_id) AS unique_customers
FROM retail_sales
GROUP BY category;
```

category	unique_customers
Beauty	141
Clothing	149
Electronics	144

Q10) Write a SQL query to create each shift and number of orders (Example Morning <12, Afternoon between 12 & 17, Evening >17)

```
SELECT
    CASE
        WHEN HOUR(sale_time) < 12 THEN 'Morning'
        WHEN HOUR(sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'
        ELSE 'Evening'
    END AS shift,
    COUNT(*) AS number_of_orders
FROM retail_sales
GROUP BY shift;
```

shift	number_of_orders
Evening	1062
Morning	548
Afternoon	377